

DATA

29-city OpenAQ panel

hourly PM_{2.5}, 14 data-rich + 15 data-scarce cities; usable windows 2.5k–19.8k h, quality-gated (§Methods)

Same cities, second domain

2-m temperature (Open-Meteo reanalysis), clipped to each city's PM_{2.5} window
→ isolates the domain effect

Beijing depth set

UCI multi-site archive: 12 stations, 2013–2017, meteorology included

Pre-registration

analysis plan locked before the campaign; every deviation logged; per-claim ledger links each number to its artifact

FIVE FORECASTING STRATEGIES

Chronos-Bolt (zero-shot)

48M-param time-series foundation model; no training, 4-week context

Chronos-Bolt + covariates

FM forecasts the residual of a ridge covariate model

LightGBM (tuned specialist)

direct multi-horizon, lag/calendar/weather features; retrained per fold

NAS-GRU (searched specialist)

2×GRU-128 from the published Green-NAS search; 5 seeds; + transfer variant (E4)

Seasonal-naïve floor

168-h persistence

EVALUATION

Rolling-origin backtest

6 folds × 24 h (48 h supplement); MASE on a fixed per-series scale

Covariate realism ablation

perfect-foresight vs causal (last-known) weather covariates — Fig. 3

Statistical rigor

per-city Diebold–Mariano (HLN, FDR); panel sign/Wilcoxon; Friedman–Nemenyi; split-conformal intervals; TOST equivalence

Measured energy → decision rule

codecarbon J per 1,000 forecasts; argmin MASE + $\lambda \cdot \text{USD}/1\text{k}$ over history × cost penalty λ — Fig. 6